A Strict Four-Row Permanent Inequality and Permutation Moments

Robust Exponent Range, Covariance Certificates, and Exact Column Laws

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Abstract

The source manuscript proves a permanent inequality below exponent two for permutation laws close to uniform with exactly uniform coordinate marginals, and inserts it into a four-row moment recursion. Using the classical sharp uniform permanent inequality, we show that every fixed exponent strictly above its uniform threshold admits such a marginal-preserving neighborhood. The threshold itself admits no positive radius. A four-permutation trade gives an explicit upper bound on the allowable radius, sharp for the single-permutation test. We supply a law-specific covariance certificate with an explicit nonlinear remainder, and classify the laws with the smallest local quadratic threshold. We also compute the source's positive four-slot column law exactly, replacing its qualitative convergence threshold by a closed total-variation formula. These statements retain the compact-complement gap, positivity, restriction multiplicities, singular-value conventions, and source recursion as separate obligations. Contribution assessment: C3/5, the full open robust exponent range, sharp single-permutation radius obstruction, and explicit local and column-law certificates refine the source's finite interfaces without independently proving the global shuffle moment theorem.

1 Normalization and the actual source claim

Write u for uniform measure on S_4. All scalar row norms and inner products use probability measure on four points:

$$\|f\|_p = \left(\frac{1}{4}\sum_{j=1}^{4}|f(j)|^p\right)^{1/p}, \qquad \langle f,g\rangle = \frac{1}{4}\sum_j f(j)g(j).$$

For a probability law ν on S_4, put

$$\mathcal{P}_\nu(f_1,f_2,f_3,f_4) = \mathbb{E}_\nu \prod_{i=1}^{4} f_i(\pi(i)), \qquad \|\nu-u\|_{\mathrm{TV}} = \frac{1}{2}\sum_{\pi\in S_4}|\nu(\pi)-1/24|.$$

The functions in a permanent inequality are real and nonnegative. Zero rows make the inequality immediate. Exact marginals mean

$$\nu\{\pi : \pi(i)=j\} = 1/4 \quad \text{for all } i,j. \tag{1}$$

The source [1] proves that some $p_0 \in (4/3, 2)$ and some $\varepsilon > 0$ satisfy

$$\mathcal{P}_\nu(f) \le \prod_i \|f_i\|_{p_0} \tag{2}$$

for every nonnegative quadruple and every law obeying (1) and $\|\nu - u\|_{\mathrm{TV}} < \varepsilon$. Its local proof uses a quadratic gap near constant rows; its global proof uses the strict exponent-two inequality away from constants.

The sharp uniform theorem of Bristiel–Caputo [2], already cited by the source, implies

$$\mathcal{P}_u(f) \le \prod_i \|f_i\|_{p_c}, \qquad p_c = \frac{4\log 4}{\log 24} \in (4/3, 2). \tag{3}$$

This is an external theorem, not proved here. The identity matrix shows necessity of $p \ge p_c$ for the uniform law. Our main result extends the source's robust *exponent range*, not the known uniform permanent threshold. Classical Taylor estimates, covariance forms, strict norm monotonicity, compactness, and finite group convolution are credited as tools, not new principles.

2 An explicit local certificate

For nonzero rows with positive means, divide out those means and write $f_i = 1 + g_i$, where $\mathbb{E}_j g_i(j) = 0$. Set $v_i = \|g_i\|_2^2$, $V = \sum_i v_i$, and $b = \max_i \|g_i\|_\infty$. On the real twelve-dimensional space $\mathcal{H} = (\{g : \mathbb{E}_j g = 0\})^4$, define the pair form

$$Q_\nu(g) = \sum_{i<j} \mathbb{E}_\nu[g_i(\pi(i)) g_j(\pi(j))], \qquad \rho_\nu = \sup_{V>0} \frac{2Q_\nu(g)}{V}.$$

It is a finite exact quadratic form for a specified law, not a covariance matrix sampled from a different distribution.

Theorem 2.1 (Law-specific remainder bound)**.** *Assume* (1)*,* $1 < p \le 2$*,* $0 \le b < 1$*, and centered rows with* $\|g_i\|_\infty \le b$*. Then*

$$\prod_i \|1 + g_i\|_p - \mathcal{P}_\nu(1 + g_1, \dots, 1 + g_4) \ge \left\{ \frac{p-1}{2(1+b)} - \frac{\rho_\nu}{2} - b - \frac{b^2}{4} \right\} V. \tag{4}$$

In particular, if $p > 1 + \rho_\nu$*, the desired inequality holds throughout an explicitly positive neighborhood of the constant rows. For* $b \le 1/2$*, it is sufficient that*

$$b \le \frac{4}{13}(p - 1 - \rho_\nu). \tag{5}$$

Proof. The four linear terms vanish by (1). The quadratic term is at most $\rho_\nu V/2$. For any triple i, j, k, Cauchy–Schwarz and the exact marginals give

$$|\mathbb{E}_\nu g_i g_j g_k| \le \frac{b}{3}(\sqrt{v_i v_j} + \sqrt{v_i v_k} + \sqrt{v_j v_k}).$$

Here g_i abbreviates $g_i(\pi(i))$. Each pair appears in two triples, and $\sum_{i<j} \sqrt{v_i v_j} \le 3V/2$, by $(\sum_i \sqrt{v_i})^2 \le 4V$. Thus all cubic terms cost at most bV. For the quartic term, bound either of the six choices of two factors by b^2, average the remaining pair bounds, and obtain $|\mathbb{E}_\nu g_1 g_2 g_3 g_4| \le b^2 V/4$.

For one centered row, the lower bound on the second derivative of x^p on $[1-b, 1+b]$ gives

$$\mathbb{E}(1+g)^p \geq 1 + \frac{p(p-1)}{2}(1+b)^{p-2}\,\|g\|_2^2\,.$$

This expectation lies between 1 and $(1+b)^p$. The derivative of $x^{1/p}$ on that interval is at least $(1/p)(1+b)^{1-p}$. Consequently

$$\|1+g\|_p \geq 1 + \frac{p-1}{2(1+b)}\,\|g\|_2^2\,.$$

Multiplying these four bounds and dropping positive cross terms proves (4). Finally $1/(1+b) \geq 1-b$, $p-1 \leq 1$, and $b^2/4 \leq b/8$ give a margin at least $(p-1-\rho_\nu)/2 - 13b/8$. This proves (5). □

Proposition 2.2 (Uniform-law and TV calibration)**.** *For the uniform law,*

$$Q_u(g) = \frac{1}{6}\left(V - \left\|\sum_i g_i\right\|_2^2\right), \qquad \rho_u = 1/3. \tag{6}$$

If ν has exact marginals and $\|\nu - u\|_{\mathrm{TV}} \leq \varepsilon$, then

$$\rho_\nu \leq 1/3 + 18\varepsilon. \tag{7}$$

Proof. Sampling two distinct positions uniformly gives $\mathbb{E}_u g_i(\pi(i))g_j(\pi(j)) = -\langle g_i, g_j\rangle/3$. Summing proves (6), with equality in its upper bound when $\sum_i g_i = 0$. For a centered function on four points, $\|g\|_\infty^2 \leq 3\,\|g\|_2^2$: at each position, $g(j) = -\sum_{k\neq j} g(k)$, then apply Cauchy–Schwarz. The TV change of one pair term is at most $2\varepsilon\,\|g_i\|_\infty\,\|g_j\|_\infty \leq 6\varepsilon\sqrt{v_i v_j}$. The total change is at most $9\varepsilon V$, proving (7). □

Example 2.3 (A rational local margin)**.** Take $p = 3/2$, $b \leq 1/32$, and $\varepsilon \leq 1/1152$, always with exact marginals. The coefficient in (4) is at least

$$\frac{8}{33} - \frac{1}{6} - \frac{1}{128} - \frac{1}{32} - \frac{1}{4096} > \frac{1}{32}.$$

This proves a strict deficit $V/32$ near constants. It is deliberately a *local* example: $3/2 < p_c$, so the global permanent inequality at this exponent fails even for u.

2.1 Physical moment coordinates

The centered Walsh basis

$$\psi_1 = (1,1,-1,-1), \quad \psi_2 = (1,-1,1,-1), \quad \psi_3 = (1,-1,-1,1)$$

is orthonormal in the probability inner product. Write $g_i = \sum_a c_{ia}\psi_a$. Let H_ν be the 12-by-12 symmetric matrix with zero diagonal blocks and off-diagonal blocks

$$(H_\nu)_{ia,jb} = \mathbb{E}_\nu[\psi_a(\pi(i))\psi_b(\pi(j))] \quad (i \neq j).$$

Then $V = c^T c$, $2Q_\nu = c^T H_\nu c$, so $\rho_\nu = \lambda_{\max}(H_\nu)$. For rational masses, H_ν is rational; a bound $\rho_\nu \leq r$ can be certified by $rI - H_\nu \succeq 0$. In nonorthonormal coordinates $c = Ra$, the condition is $R^T(rI - H_\nu)R \succeq 0$, with metric $R^T R$. An ordinary Euclidean eigenvalue of $R^T H_\nu R$ is not the physical threshold. This exact metric transport is the relevant Ram moment synthesis; changing coordinates alone is not an improvement.

3 The smallest local threshold and its limits

Theorem 3.1 (Classification at the quadratic minimum)**.** *Every law satisfying* (1) *has* $\rho_\nu \geq 1/3$. *Equality holds if and only if*

$$\nu(\pi) = \frac{1 + \tau \operatorname{sgn} \pi}{24}, \qquad -1 \leq \tau \leq 1. \tag{8}$$

For a fixed law, $p-1 \geq \rho_\nu$ *is necessary for a permanent inequality on a full neighborhood of constants, and* $p - 1 > \rho_\nu$ *is sufficient. No assertion at equality is needed.*

Proof. Let $C = I + H_\nu$, the full covariance matrix of the twelve variables. For a common centered row $g_1 = \cdots = g_4 = h$, $\sum_i h(\pi(i)) = \sum_j h(j) = 0$ pointwise. Hence C vanishes on the three-dimensional common-row subspace, and $H_\nu = -I$ there. Since $\operatorname{Tr} H_\nu = 0$, the other nine eigenvalues sum to three. Their largest is at least $1/3$. If the largest is $1/3$, all nine are $1/3$; therefore

$$H_\nu = \frac{I_4 - J_4}{3} \otimes I_3.$$

Its off-diagonal pair form agrees with uniform sampling without replacement. Together with the uniform one-coordinate marginals, the complete basis $1, \psi_1, \psi_2, \psi_3$ implies that every two-coordinate joint law is uniform on the twelve distinct pairs.

Fix any permutation π and any transposition s of two input positions. Prescribing the images of the other two positions leaves exactly the completions $\pi, \pi s$. Pairwise uniformity thus gives $\nu(\pi) + \nu(\pi s) = 1/12$. For $a(\pi) = \nu(\pi) - 1/24$, this says $a(\pi s) = -a(\pi)$. Transpositions generate S_4, so $a(\pi) = a(e) \operatorname{sgn} \pi$. Nonnegativity gives (8). Conversely, its sign contribution cancels between the two completions of any two prescribed images; all pair laws are uniform and $\rho_\nu = 1/3$.

For the final assertion, replace g by tg, $t \downarrow 0$. The second-order deficit is $\frac{1}{2}t^2[(p-1)V - 2Q_\nu(g)]$. If $p - 1 < \rho_\nu$, choose a maximizing vector and small positive t, making the deficit negative while every row remains positive. Sufficiency follows from Theorem 2.1. □

For (8), the exact permanent functional is

$$\mathcal{P}_\nu(f) = \frac{\operatorname{perm}(f_i(j)) + \tau \det(f_i(j))}{24}.$$

The determinant contributes no quadratic term near the constant matrix. For centered rows its degree-four part vanishes because the centered row space has dimension three; only terms of degree three can remain. Nevertheless the identity-row test, choosing a permutation of parity matching τ, imposes

$$\frac{1 + |\tau|}{24} \leq 4^{-4/p}.$$

For $|\tau| \geq 1/2$, every $p < 2$ fails globally although the strict local inequality holds for every $p > 4/3$. Thus even the exact minimal covariance threshold is not a certificate of the source's all-row inequality.

4 The full open robust exponent range

Theorem 4.1 (Robustness above the sharp uniform threshold)**.** *For every fixed* $p \in (p_c, 2)$*, there is* $\varepsilon_p > 0$ *such that* (2)*, with exponent* p*, holds for all nonnegative rows and all laws satisfying* (1) *and* $\|\nu - u\|_{\mathrm{TV}} < \varepsilon_p$*. No positive such radius exists at* $p = p_c$*, and for* $p < p_c$ *even the uniform law fails.*

Proof. Zero rows are immediate; normalize every other row by $\|f_i\|_2 = 1$. These nonnegative rows form a compact space $\mathcal{S}$; each entry is at most two. Put $\delta = p - 4/3 > 0$, $b_p = \delta/13$. Let $\mathcal{U}$ consist of the normalized quadruples whose mean-normalized centered rows have maximum norm less than b_p. All row means are positive, so this is an open neighborhood of the constant quadruple.

If $\varepsilon \le \delta/36$, then $\rho_\nu \le 1/3+\delta/2$. The proof of Theorem 2.1 gives throughout $\mathcal{U}$ a mean-normalized deficit at least $\delta V/8$. Homogeneity returns the inequality to the original rows.

On $\mathcal{S} \setminus \mathcal{U}$, at least one row is nonconstant. Strict monotonicity of probability L^p norms gives

$$\mathcal{P}_u(f) \le \prod_i \|f_i\|_{p_c} < \prod_i \|f_i\|_p .$$

The first inequality is the imported sharp uniform theorem (3); no equality classification at p_c is needed. By compactness,

$$\gamma_p := \min_{\mathcal{S}\setminus\mathcal{U}} \left(\prod_i \|f_i\|_p - \mathcal{P}_u(f) \right) > 0. \tag{9}$$

Since the product of four entries is at most sixteen, perturbing the law changes the functional by at most 32ε. Thus any

$$0 < \varepsilon_p \le \min\{\delta/36, \gamma_p/32\} \tag{10}$$

works, with strict TV inequality. The failure at and below the threshold follows from the next proposition and the identity test. □

The radius in (10) is not fully numerical: the compact-complement minimum γ_p has not been evaluated. This theorem strengthens the source's existential exponent near two to every exponent in the stated open interval, using a stronger classical uniform input already identified in the source. It does not assert historical priority for this robustness corollary.

Proposition 4.2 (Sharp single-permutation obstruction)**.** *For exact-marginal laws and* $0 \le \varepsilon \le 1/12$*,*

$$\max_{\|\nu-u\|_{\mathrm{TV}} \le \varepsilon} \nu(e) = \frac{1}{24} + \frac{\varepsilon}{2}.$$

Consequently any radius on which the permanent inequality with $p \in [p_c, 2)$ *holds for every exact-marginal law must satisfy*

$$\varepsilon_p \le 2\left(4^{-4/p} - \frac{1}{24}\right). \tag{11}$$

This upper bound is sharp for this test, not claimed sharp for the full permanent inequality.

Proof. Write $\Delta = \nu - u$, and let $a = \Delta(e) > 0$. Exact marginals give zero change in the expected number of fixed points. The positive part of Δ contributes at least $4a$. Every nonidentity permutation has at most two fixed points, and the negative part excludes e, so its contribution is at most $2\|\nu - u\|_{\mathrm{TV}}$. Hence $a \le \varepsilon/2$.

For attainment, let $s = (12)$, $t = (34)$ and

$$\nu_a = u + a(\delta_e + \delta_{st} - \delta_s - \delta_t), \qquad 0 \le a \le 1/24.$$

The signed measure has mass zero and every coordinate-image marginal zero, as may be checked separately on the two disjoint pairs. Its TV norm is $2a$, it preserves nonnegativity, and $\nu_a(e) = 1/24+a$. This attains the upper bound. Choose $f_i(j) = 4^{1/p}\mathbf{1}_{\{j=i\}}$; each has norm one, and the left side is $4^{4/p}\nu_a(e)$. It exceeds one whenever $a > 4^{-4/p} - 1/24$. For $p \in [p_c, 2)$ the critical a lies in $[0, 1/48)$, within the permitted range. Arbitrarily small positive a defeats the endpoint $p = p_c$. □

5 The column law can be computed exactly

This section concerns the *positive column operator* M^*M, not the forward shuffle. Use slots $1, 2, 3, 4$ and

$$H = \langle (12), (34) \rangle, \qquad K = \langle (13), (24) \rangle.$$

Let $A = U_H$, $B = U_K$ in the group algebra, so the source's two-axis column operator is $M = BA$, and $M^*M = ABA$. Let D be the order-eight subgroup preserving the unordered partition $\{\{1, 2\}, \{3, 4\}\}$.

Theorem 5.1 (Exact positive column mixing)**.** *For every integer $k \geq 1$, the law of $(M^*M)^k$ is*

$$\omega_k = (1 - 4^{-k})u + 4^{-k}U_D, \qquad \|\omega_k - u\|_{\mathrm{TV}} = \frac{2}{3}\,4^{-k}. \tag{12}$$

Its one-coordinate marginals are exactly uniform. In the regular representation, M^*M has eigenvalues $1, 1/4, 0$ of multiplicities $1, 2, 21$, respectively.

Proof. The subgroup K consists of $e, (13), (24), (13)(24)$. The last double transposition normalizes H, and together with H generates D. Thus averaging HkH for $k = e, (13)(24)$, each with its weight $1/4$ in K, contributes $\frac{1}{2}U_D$. For $k = (13)$ or (24), the intersection $H \cap kHk^{-1}$ is trivial. Each double coset HkH has sixteen elements and is disjoint from D, hence equals $S_4 \setminus D$; its convolution law is uniform there. These two elements of K contribute $\frac{1}{2}U_{S_4 \setminus D}$. Therefore

$$ABA = \tfrac{1}{2}U_D + \tfrac{1}{2}U_{S_4 \setminus D} = \tfrac{3}{4}u + \tfrac{1}{4}U_D.$$

Uniform subgroup averaging is idempotent, and u absorbs every probability law. Taking powers proves the first formula. The TV distance between U_D and u is $1 - 8/24 = 2/3$. The group D is transitive on the four slots, so both mixture components have uniform marginals. In the regular representation U_D is an orthogonal projection of rank $[S_4 : D] = 3$, while u has rank one and its range is contained in that of U_D. The spectral assertion follows. □

In the source $k = r/2$, with r even. For any available robust radius ε_p, the column-law requirement is therefore exactly

$$\frac{2}{3}\,4^{-r/2} < \varepsilon_p.$$

One may choose the smallest integer $k \geq 1$ satisfying this inequality and take $r = 2k$, before enlarging r to meet the *other* sparse-stopping and finite-size requirements of the source. This computes one threshold, not the final source moment order: γ_p, the tree cutoff and finite-size spectral gates remain. For example $k = 4$ gives TV $1/384$, and $k = 5$ gives $1/1536 < 1/1152$, so the rational local example applies to the latter column law. Its local exponent $3/2$ must still not be used as a global recursion exponent.

6 What the recursion still requires

The local result is not a substitute for the source's representation argument. Here we record its actual laws and the positivity that makes the permanent input usable. Let $N = 2^d = 4m$, let T_N be the averaged ordered coordinate sweep in a unitary representation, and put $B_N = T_N T_N^*$. This is a positive contraction. One sweep comprises d physical shuffles. Neither T_N nor its powers need be normal. For a Specht module of dimension D_λ, the source studies

$$F_\lambda = D_\lambda \operatorname{Tr} B_N^r, \qquad r \text{ even}.$$

The first $d-2$ coordinate levels act within four rows of size m. If $A = B_m^{\otimes 4}$ and M is the last two-level column operator, then $B_N = MAM^*$. With $X = A^{r/2}$ and $Y = (M^*M)^{r/2}$, the Araki–Lieb–Thirring inequality gives the source reduction

$$\operatorname{Tr} B_N^r \le \operatorname{Tr}(XYXY) = \left\| Y^{1/2} X Y^{1/2} \right\|_{\mathrm{HS}}^2 .$$

Both positivity assumptions are essential. Splitting X into row isotypes means using the entire central isotypic projection, including every multiplicity copy, before applying the Hilbert–Schmidt triangle inequality. Selecting one copy would not bound the original trace.

6.1 Why the coloring operator is positive

For an irreducible unitary representation ρ of a finite group G, of dimension D, and $0 \neq Q \ge 0$, the source's squared coefficient law is

$$u_Q(g) = \frac{D}{|G| \operatorname{Tr} Q^2} |\operatorname{Tr}(Q\rho(g))|^2. \tag{13}$$

Schur orthogonality makes this a probability law. The function $\operatorname{Tr}(Q\rho(g))$ is positive definite: diagonalizing Q expresses it as a nonnegative sum of matrix coefficients $\langle v, \rho(g)v\rangle$. Its absolute square is positive definite by the tensor-product representation. Hence convolution by u_Q is positive in every unitary representation. It is also inversion invariant. In particular the induced operator U on the 4^m row colorings is positive, symmetric, stochastic and entrywise nonnegative. An arbitrary inversion-invariant probability law does not have this property: on the group of order two, mass one at the nonidentity element gives eigenvalue -1.

For such a positive stochastic matrix of size s, and $1 < p < 2$, ordinary interpolation gives

$$\sum_{x,y} U(x,y)^p \le s^{\,2-p} (\operatorname{Tr} U^2)^{p-1} \le s^{\,2-p} (\operatorname{Tr} U)^{p-1}. \tag{14}$$

Indeed apply Hölder to the entrywise identity $U^p = U^{2-p}(U^2)^{p-1}$; use $\sum U = s$ and $\sum U(x,y)^2 = \operatorname{Tr} U^2$. Finally all eigenvalues are in $[0,1]$. Keeping $\operatorname{Tr} U^2$ is a valid sharper finite bound, but no new uniform multiplicity estimate for it is claimed here. The normalized entrywise norm is consequently at most $s^{-1}(\operatorname{Tr} U)^\theta$, where $\theta = 1 - 1/p < 1/2$.

The two colorings in the compatibility event come from independent column laws on the left and right. Within one column their four colors form a permutation, not four independent colors. Tensorization therefore uses the *global* four-row permanent inequality for each column, not merely the covariance certificate near constants. With $s = 4^m$, the resulting factor s^{-4} cancels the crude multinomial bound $N!/(m!)^4 \le 4^N = s^4$. This is the concrete place where the exponent is used.

6.2 Multiplicity and stopping costs

In the source, $\operatorname{Tr} U_\alpha$ is bounded by

$$C_\alpha = \sum_{\substack{n_1+\cdots+n_4=m \\ n_i \ge 0}} \sum_{\beta_i \vdash n_i} \left(c^\alpha_{\beta_1,\beta_2,\beta_3,\beta_4}\right)^2,$$

where $c^\alpha_{\boldsymbol{\beta}}$ is the restriction multiplicity to the four corresponding symmetric groups. The resulting recurrence has the form

$$F_\lambda^{1/2} \le \sum_{\boldsymbol{\alpha} \text{ occurring}} \prod_{i=1}^{4} (F_{\alpha_i} C^\theta_{\alpha_i})^{1/2}. \tag{15}$$

The occurrence restriction and the squares of multiplicities matter. The source obtains a multiplicity-square estimate using both restriction and induction dimension inequalities, then removes rows and columns by Pieri's rule. The tail budgets must be attached to constituents that actually occur in the original restriction. A rank bound on an auxiliary matrix does not reproduce this argument.

The following is the complete scientific module map, not a claim that the global proof was independently formalized.

Module	Mechanism and retained obligation
01 introduction	Fixed even moment and representation-dimension decay; ordered-sweep convention and support lower bound.
02 permanent	Local Taylor gap, compact-complement strictness, independent-column tensorization and positive trace interpolation.
03 recursion	ALT reduction, whole isotypes, squared Fourier coefficients, two independent colorings and exact compatibility law.
04 multiplicities	Character domination, restriction squares, Pieri and Mackey counts, hook-deficit budgets for occurring tuples.
05 sparse	Ordered tagged injections, shared-switch coupling, alternating cancellation, encounter-forest estimates, sparse and finite-size stopping.
06 tree	Levelwise choice and multiplicity costs, including stopped children; comparison with hook-formula dimensions, in a fixed order of parameter selection.
07 amplification	Singular-value control for powers of a possibly nonnormal sweep, reciprocal dimension summation, then Plancherel.

In more detail, the sparse module couples all tagged subsets through the same actual switch realization. Binomial inversion removes encounter patterns that fail to cover all tags. Its forest estimate, valid in its stated sparse regime, is not a claim that the full permutation law is independent. The tree module expands nonsparse large nodes and sums both choice costs and multiplicity costs; stopping a child does not erase its C_α^θ cost. These global estimates, and the source's fixed finite-size spectral cutoff, are imported here. Computing the four-slot law does not independently establish them.

The benefit of using p closer to p_c is a smaller $\theta = 1 - 1/p$, hence potentially smaller multiplicity costs. But its robust radius may shrink; the unknown γ_p prevents an explicit optimized final moment order. No improved physical mixing constant is asserted.

6.3 Nonnormal amplification and the physical clock

If the full source recurrence has already proved $D \operatorname{Tr}(TT^*)^r \le D^{1-\eta}$, then $\|T\|_{\mathrm{op}}^{2r} \le D^{-\eta}$. For integer $q \ge r$, Schatten Hölder gives

$$\|T^q\|_{\mathrm{HS}}^2 \le \|T\|_{\mathrm{op}}^{2(q-r)} \, \|T\|_{2r}^{2r} \le D^{-\eta q/r}.$$

Here the first inequality follows by treating $q - r$ factors in operator norm and r factors in Schatten norm $2r$. Thus $D \, \|T^q\|_{\mathrm{HS}}^2 \le D^{1-\eta q/r}$. The remaining dimension sum and sign representation must still be handled before total variation follows. It would be incorrect to replace $T^q(T^*)^q$ by $(TT^*)^q$ without normality. These are q coordinate sweeps, or qd physical shuffles, not q individual shuffles.

7 Original Ram proofs and finite certification

The original TR250 passages [6] used here distinguish physical quadratic forms from coordinate descriptions. A finite moment bank $m = Ec$ admits a left inverse only on its range when E has

full column rank. An off-range input requires a specified projection metric. Its polynomial Gram argument uses positivity of the integral of the square of a nonzero polynomial; positive quadrature weights alone do not guarantee that sampled polynomials are distinguishable. Applied here, these facts justify the physical form $\sum_i \|g_i\|_2^2$ and the congruence test in Section 2, not a replacement of the law ν by arbitrary sampled moments. The twelve-dimensional bank is exact because the rows are functions on four points.

The original clearance chapter [7] proves the classical Bernstein convex-hull certificate and its exact rational subdivision rule. Here it can certify a *whole interval* of local radii. For example, set $p = 3/2$ and $\rho = 1/3 + 1/64$, as in the rational local example. Multiplying the proposed lower margin $1/32$ by $4(1+b) > 0$ reduces it to positivity of the cubic

$$R(b) = 2(p-1) - 2\rho(1+b) - 4b(1+b) - b^2(1+b) - \tfrac{1}{8}(1+b). \tag{16}$$

On $b = x/32$, $0 \le x \le 1$, write $R(x/32) = \sum_{j=0}^{3} a_j x^j$. Its degree-three Bernstein coefficients are exactly

$$\beta_j = \sum_{k=0}^{j} a_k \frac{\binom{j}{k}}{\binom{3}{k}}.$$

The four coefficients are

$$(\beta_0, \beta_1, \beta_2, \beta_3) = \left(\frac{17}{96}, \frac{1169}{9216}, \frac{691}{9216}, \frac{703}{32768}\right) > 0.$$

Since $\binom{3}{j} x^j (1-x)^{3-j} \ge 0$ and their sum is one, this certifies the full interval $0 \le b \le 1/32$, not a finite grid. The identity for β_j follows by expanding $x^k = x^k(x + (1-x))^{3-k}$. No global nonlinear permanent inequality follows from this local cubic certificate. In particular, the compact-complement gap γ_p has not been evaluated by subdivision.

These original proofs supply the representation discipline and a concrete local polynomial certificate. They do not supply the sharp uniform permanent theorem, the new exponent-range deduction, or the source's all-size recursion. Those dependencies are separate. The present work also differs from M476: no complex analytic continuation or Green-potential estimate is used, and no result from that report is counted as a fresh contribution.

8 Evidence, status, and limitations

All ten cached public text files were freshly displayed in bounded, untruncated reads: README (16 lines), main (48), bibliography (166), and the seven scientific modules (68, 84, 176, 151, 243, 141 and 43 lines). Thus the scientific read is 906 lines and the full text read is 1,136 lines. The source byte-identity record compares captured trees; it does not establish current branch state or mathematical validity. The original private proof reads total 193 lines: TR250, lines 121–181 and 425–477; clearance, lines 62–140. The associated ledger records hashes captured before these reads, post-read hashes, and all unread complements. No whole-corpus or complete-private-file read is claimed.

The public refresh on October 9, 2026 checked the primary Bristiel–Caputo statement and its normalization passage (Corollary 1.14 and Section 4.3), the Carlen–Lieb–Loss theorem statement, and the ALT statement in Audenaert's exposition. The full external proofs were not independently audited. The publisher's family-238 scope page [5] describes the selected formalized result for this manuscript as the robust four-row permanent inequality, and expressly excludes its Thorp mixing

consequence from that selected statement. This is publisher scope metadata, not a Lean execution or a proof audit performed here. No historical priority is claimed for the robust corollary.

The owned test script enumerates finite group laws, exact marginal constraints, covariance matrices, polynomial coefficients and the column convolution. The final run passed 291 parameter cases and 7,126 assertions using exact rational or symbolic rational arithmetic, with no floating-point diagnostics. This includes eight column-power cases, nine parity laws, 25 trades, 96 translated trades, 120 centered polynomial fixtures, six physical-coordinate congruences, and the matrix, constraint-kernel, fixed-point and Bernstein checks. The receipt separates cases from assertion evaluations; parent-agent checks are not included. These tests supplement the proofs; they do not compute γ_p, certify arbitrary representation sizes, or replace the imported Bristiel–Caputo theorem.

8.1 Advantages and costs

The exponent theorem identifies the entire open interval above the known sharp threshold, rather than an unspecified exponent near two. The obstruction shows why a positive robust radius cannot persist at the threshold. The covariance certificate is inexpensive for an explicit 24-atom law and isolates precisely the necessity of exact marginals. The parity family shows that the best quadratic behavior can coexist with bad global behavior. The exact column mixture removes one qualitative convergence limit from the recursion.

The cost is important: the sufficient global radius still contains a compact minimization, whereas the explicit radius obstruction is only an upper bound. Bernstein certification of the local cubic is not certification of this compact nonlinear problem. A small local matrix does not reduce the dimension-dependent source representation proof to finite enumeration. None of the finite calculations measures an implementation speedup.

8.2 Extensions that still need work

One useful next problem is a certified lower estimate for γ_p, preferably tracking its degeneration as $p \downarrow p_c$. This would turn the exact column-law formula into an explicit input for the source parameter selection, though sparse and tree constants would remain. Another is to characterize the exact global robustness radius: the single-permutation trade need not be the worst test. One may also retain $\operatorname{Tr} U^2$ in (14), but a useful global gain would require new uniform representation estimates compatible with the same restriction and stopping budgets. These are research directions, not proved mixing improvements.

9 Conclusion

The demonstrated increment is finite and source-specific: the full open robust exponent range, a sharp single-permutation TV obstruction, an explicit nonlinear local certificate and its extremal covariance classification, and the exact positive column law used in the source. The arguments use classical permanent, norm, positivity and compactness tools with explicit attribution. They neither settle a new uniform permanent threshold nor independently prove the global Thorp moment or mixing theorem. This supports C3 for the local quantitative and structural improvements, independently of the catalogue's L3 applicability assessment.

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